

Professional sports teams are now combining wearables with radio frequency identification (RFID) technology. RFID tags added to players' uniforms transmit data using electromagnetic fields to RFID receivers placed at various locations around the arena or stadium.

Tags collect information about players' movements such as speed, acceleration and distance run. Coaches can use this information to get a better sense of how their players are performing and working together on the field. These stats are sometimes even shown in broadcasts or on Jumbotron, so fans can see where their favourite players stack up.

Fitness electronics

To be a good sports person, one must have a keen interest in new technologies that allow him or her to understand more about how the human body copes with various training methodologies. In order to improve the quality of training prescriptions, one needs to base this on data and be able to provide an evidence based approach to athletic training.

We now have access to sophisticated tools that are capable of measuring different aspects of human performance; mainly related to the output of movement, for example, power output, force and velocity.

Electronics and computers incorporated into clothing or accessories that can be worn on the body are now being designed for use in sports and fitness activities. These wearable technologies gather data such as heart rate, breathing rate, skin temperature, calories burned, movement and even the duration and quality of sleep.

Collected data is analysed by software in conjunction with inputted information about the wearer's height, weight and gender. The wearer can then track and view this information on a computer or mobile device and use it to make better decisions about health, fitness or athletic performance.

One major trend in sports technology is to get more accurate data from athletes. Sports technology has made a giant leap forwards over the past year. New devices now connect using Bluetooth 4.0 for low-power usage. Better data analytics means having more information about a workout of a sports person, and many of these gadgets offer easily-adjustable, customisable features to fit the fitness regimen.

Fitness bands like Fitbit are wearables equipped with a pedometer containing micro-electro-mechanical systems (MEMSes) inertial sensors to detect steps.

A smart tech polo shirt has also been unveiled, which includes silver-fibre sensors woven onto the fabric, an accelerometer (which turns bodily movements, or acceleration, into digital measurements), a gyroscope (which measures orientation) and Bluetooth transmitter to relay information about wearers' vitals.

One of the best ways to keep tabs for a sportsman on his workout is to use a fitness watch that can monitor heart rate and track distances, speeds and interval times, and send this information to an online system that helps to develop a personal training regimen that will give the best results.

A standard watch with a basic chronograph (stopwatch) and heart rate monitor gives all the health-related information needed to create a training schedule. When choosing a heart rate monitor, the one that can give customised information based on height, weight, age and other factors works well. These watches work great for training programmes that include running, walking, cycling or weight lifting.



Implantable electronics (Image courtesy: cloudwiser.wordpress.com)

Outdoor technology has advanced so much that many fitness watches come with GPS technologies as well as heart rate monitoring capabilities. GPS can be used to track location, distance travelled, top and average speeds, pace and other data on workout. These watches are great for measuring performance on runs, walks and bike rides.

Implantable electronics

Portable ultrasound and electrostimulation therapy equipment are being used in sports medicine, physical therapy, chiropractic and podiatric health care.

Further, portable and field biochemistry technology has improved massively but, still, a part from blood lactate, glucose and a few other biomarkers are quite limited in the ability for measuring a wide variety of biomarkers *in-vivo*.

Researchers are using silk as the basis for implantable optical and electronic devices that are capable of measuring vital signs and blood biochemistry in real-time in a continuous manner. These implantable electronics are based on silk, which is a biodegradable material and is capable of carrying light like optical glass.

Silk can also serve as mechanical support for arrays of electrically-active devices, allowing these to be placed on biological tissues without causing irritation.